

Submission Worksheet

IT490-450-M2026 / mq-practice-individual

Submission Data

Course	IT490-450-M2026
Assignment	MQ Practice Individual
Student	Ruchir P. (rkp28)
Status	submitted
Worksheet Progress	100%
Potential Grade	10.00/10.00 (100.00%)
Received Grade	0.00/10.00 (0.00%)
Started	6/14/2026, 10:04:19 PM
Updated	6/14/2026, 10:54:09 PM

Grading Link

<https://courses.ethereallab.app/assignment/v3/IT490-450-M2026/mq-practice-individual/grading/rkp28>

View Link

<https://courses.ethereallab.app/assignment/v3/IT490-450-M2026/mq-practice-individual/rkp28>

Instructions

1. Read all instructions and requirements first
2. Use any VM creation tool that gives you root access and persistent storage
 - VirtualBox, Multipass, cloud (Amazon, Google, Azure, etc) (Docker won't be an option here)
 - Create a hostname relevant to the assignment (i.e., test-individual)
 - Create a user of your ucid with a password, ensure relevant permissions
 - Hardware: 1GB Memory, 10GB Hard Drive
 - Install a server version of linux (i.e., Ubuntu Server 24.04)
 - **Hint:** You may want to get a base install working and use that as a cloning point for quicker destroy/create cycles
3. Get the sample code from the **main** branch at <https://github.com/MattToegel/IT490-2026>
4. Connect to one VM with two separate ssh connections
 - Run the RabbitMQServerSample.php file successfully in one instance
 - Run the RabbitMQClientSample.php file successfully in another instance
 - Proper data should be sent/received
5. Create a **setup.sh** script that automates the installation/setup logic to get this sample working (steps needed after the installation and login of Ubuntu)
6. Create a branch in the format of **module02-<ucid>-individual-practice** in your team's repository (provided by the instructor)
7. Upload the content of the **setup.sh** script to it and the composer.json
8. Fill in the below requirements
9. Submit and Export once done
10. Upload the PDF to the branch you created above in your group GitHub repo for the class
11. No need to merge (Pull Request) this branch
12. Upload the PDF to Canvas
13. You may delete any test VMs created for this assignment

1. You need to merge (or fork) this branch
12. Upload the PDF to Canvas
13. You may delete any test VMs created for this assignment

Submission Progress

100%

Section #1: (7 pts.) Example Solution

100%

Task #1 (3.5 pts.) – Demonstrate successful example

100%

Details

- Run the echo example and capture both the server and client output
- Run the ping/pong example and capture both the server and client output
- UCID (user) must be present
- Clearly caption each image

```
root@kali:~/code# ./test.php --server=127.0.0.1 --port=8080 --mode=echo
[2024-06-10 14:00:00] INFO: Server started on 127.0.0.1:8080
[2024-06-10 14:00:01] INFO: Client connected from 127.0.0.1:54321
[2024-06-10 14:00:01] INFO: Echoing message: test message
[2024-06-10 14:00:01] INFO: Message received: test message
[2024-06-10 14:00:01] INFO: Client disconnected
[2024-06-10 14:00:01] INFO: Server stopped
```

test message

```
root@kali:~/code# ./test.php --server=127.0.0.1 --port=8080 --mode=pingpong
[2024-06-10 14:00:00] INFO: Server started on 127.0.0.1:8080
[2024-06-10 14:00:01] INFO: Client connected from 127.0.0.1:54322
[2024-06-10 14:00:01] INFO: Ping message received: pong
[2024-06-10 14:00:01] INFO: Pong message received: ping
[2024-06-10 14:00:01] INFO: Client disconnected
[2024-06-10 14:00:01] INFO: Server stopped
```

ping message test

Task #2 (3.5 pts.) – Setup Script

100%

Details

- This script must do all of the install/setup steps in order to immediately test the example code
 - Installing any tools needed (i.e., RabbitMQ server, etc)
 - Downloading the example code from my GitHub (including unzipping if necessary for your workflow)
 - Any PHP/Composer processes
- The script should be able to be executed multiple times without having or causing an error
- Setup the composer.json to include a script section to be able to invoke this script via the **composer** command

Combo #1

Part 1: Images 100%

Details

- Show the content of the **setup.sh** script you created to automate the installation and configuration steps that lead up to a working example.
 - Include comments noting what each line is doing
 - Comments should contain your ucid
- Show the raw output of the script being freshly executed successfully (may be best to pipe it to a file and capture the content)
- Include a screenshot of the modified **composer.json**
- Clearly caption each image

```
root@kali:~/code# ./setup.sh
[2024-06-10 14:00:00] INFO: Installing dependencies
[2024-06-10 14:00:01] INFO: Downloading example code
[2024-06-10 14:00:01] INFO: Unzipping example code
[2024-06-10 14:00:01] INFO: Installing RabbitMQ
[2024-06-10 14:00:01] INFO: Starting RabbitMQ
[2024-06-10 14:00:01] INFO: Server ready for testing
```

```
root@kali:~/code# ./test.php --server=127.0.0.1 --port=8080 --mode=echo
[2024-06-10 14:00:00] INFO: Server started on 127.0.0.1:8080
[2024-06-10 14:00:01] INFO: Client connected from 127.0.0.1:54323
[2024-06-10 14:00:01] INFO: Echoing message: test message
[2024-06-10 14:00:01] INFO: Message received: test message
[2024-06-10 14:00:01] INFO: Client disconnected
[2024-06-10 14:00:01] INFO: Server stopped
```

```
#!/bin/bash
# This script sets up the environment for the course.
# It installs the necessary tools and sets up the workspace.

# Install the necessary tools
sudo apt-get update
sudo apt-get install -y git curl wget unzip

# Set up the workspace
mkdir -p ~/workspace
cd ~/workspace

# Clone the repository
git clone https://github.com/MattToegel/it490-m2026-racingdevs.git
cd it490-m2026-racingdevs
```

script contents

```
#!/bin/bash
# This script runs the setup script.
# It sets up the environment for the course.

# Run the setup script
./setup.sh
```

script running

Part 2: URLs 100%

Details

- Include a link to the `setup.sh` script from your personal branch on the group repository
- Include a direct link to your modified `composer.json`

URL #1

<https://github.com/MattToegel/it490-m2026-racingdevs/blob/module02-rkp28-individual-practice/setup.sh>

URL #2

<https://github.com/MattToegel/it490-m2026-racingdevs/blob/module02-rkp28-individual-practice/composer.json>

Section #2: (3 pts.) Reflection

100%

Task #1 (1 pt.) – What was the easiest part of this assignment

100%

Details

- At least a few thoughtful sentences

Response

The easiest part for me was setting up the VM due to the help and set up from the professor. It was easy to SSH into both terminals and

Supports ****bold****, **italic**, [links](url), `code`, etc.

136 chars

Task #2 (1 pt.) – What was the hardest part of this assignment

100%

Details

- At least a few thoughtful sentences

Response

The hardest part for me was just getting vendor installed I was having PHP download issues.

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92 chars

Supports **bold**, *italic*, [links](url), `code`, etc.

92 chars

Task #3 (1 pt.) – What did you learn during this assignment

100%

Details

- At least a few thoughtful sentences

Response

I learned how to fix the vendor issue and also learned how to push and commit stuff on git.

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92 chars

End of Assignment — make sure you click **Submit**.